

Off-the-job apply tasks - Introduction to data quality – Transforming low-quality datasets

The brief

In this exercise, your task is to create a high-quality dataset by consolidating data from three separate low-quality datasets. These are currently provided as three separate sheets in the file you can find in the Hub next to this briefing document:

[Mwanza r, Pwani r, DSM\(ilala\) r](#)

In this scenario, the data science team have asked you, a data engineer, to investigate the issues with the available data and to make it usable for data scientists. You have received the following brief in an email:

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From: <datascientist@company.com>
To: <dataengineer@company.com>
Re: Dataset usability - healthcare facility data

Good morning, as part of our new healthcare facility project for Tanzania, we would like to perform clustering on the data for the three localities Mwanza, Pwani and Ilala. The data has been sourced from local authorities, they all use slightly different formats. Can you please pool this into a single dataset and make sure any quality issues are addressed. Please also create a schema for the data.

Many thanks,
```

When working on this exercise, you will find the following considerations and questions helpful:

- You may want to drop (skip) the SRA Code as it's only provided for one dataset and not useful for the other two (i.e. the majority of) datasets. Make sure you document your choice
- What health facility types are there? (Hint: you can find out from the header in the Mwanza sheet)
- What ownership types are there? You may want to include ownership type as it's useful for the majority of the datasets (two datasets specify it, one doesn't). Make sure you document any assumptions you make as to the meaning.
- Are star levels / rating levels the same for all the datasets? Is it safe to assume that they may follow the same distribution or not?
- Pwani and DSM(Ilala) contain multiple tables inside them, so you need to merge the data. You may want to skip the row numbering as it's not particularly useful at this stage. Note how in DSM(Ilala) the tables are "side-

by-side" and need to be "split in half" and there is one missing column name across all of them.

- Pwani does not explicitly declare facility types but they can be inferred from the facility name
- Pay attention to spelling mistakes, duplicates and missing data and have a plan of how to deal with those. Make sure the capitalisation remains consistent
- Create a schema with at least the following fields: Region, HF Name, HF Type, Ownership Type. Decide what to do with star rating. In the schema, explain the data types, ranges, whether missing values are allowed, and provide a description of the meanings
- The schema should be in the same excel file as your clean data, albeit as a separate sheet

To finish:

Now document all your steps. Draft an email response to the Data Scientists, including your clean usable data, and all the steps and assumptions you made.